

This writeup aims to explain basis splines, or more commonly known, B-splines. The basic idea is that they (exactly) represent every space of piecewise polynomial functions. Furthermore, linear combinations of splines are also known to universally approximate any continuous function on a compact interval. Here, I follow the standard reference by Carl de Boor (1978).

1 B-Splines

First I provide an important definition.

Definition 1.1 (k^{th} divided difference). *The k^{th} divided difference of a function g at the points $\tau_i, \dots, \tau_{i+k}$ is the leading coefficient of the polynomial of order $k + 1$ which agrees with g at the points $\tau_i, \dots, \tau_{i+k}$. It is denoted by*

$$[\tau_i, \dots, \tau_{i+k}] g.$$

One can think of interpreting this as the discrete k^{th} derivative.

Example 1. Let $g(x) = x^2$, $\tau = (0,1,3)$, what is $[0 \ 1 \ 3]g$? Following the definition, consider the polynomial of order 3, $p(x) = ax^2 + bx + c$, and evaluate at the points $g(0) = 0, g(1) = 1, g(3) = 9$. These are given by $p(0) = 0, p(1) = a + b, p(3) = 9a + 3b$, so solving these equations we have $a = 1, b = 0$. Conclude that $[0 \ 1 \ 3]g = 1$.

The following is the standard definition of B-splines.

Definition 1.2 (B-splines). *Let $\mathbf{t} := (t_i)_i$ be an increasing sequence. The i^{th} B-spline of order k for the knot sequence $\mathbf{t}$ is denoted by $B_{i,k,\mathbf{t}}$ and is*

$$B_{i,k,\mathbf{t}}(x) := (t_{i+k} - t_i)[t_i, \dots, t_{i+k}](\cdot - x)_+^{k-1}$$

We often write B_i instead of $B_{i,k,\mathbf{t}}$ to reduce notation.

Example 2. Consider a B-spline with knot index $i = 0$, order $k = 2$, and knot sequence $\mathbf{t} = (0,1,2)$. Then $B_{i,k,\mathbf{t}}(x) := (t_{i+k} - t_i)[t_i, \dots, t_{i+k}](\cdot - x)_+^{k-1} = 2[0, 1, 2](\cdot - x)_+$. One can solve $B_{0,2,\mathbf{t}}(x) = (2 - x)_+ - 2(1 - x)_+ + (-x)_+$.

Here are some properties of B-splines.

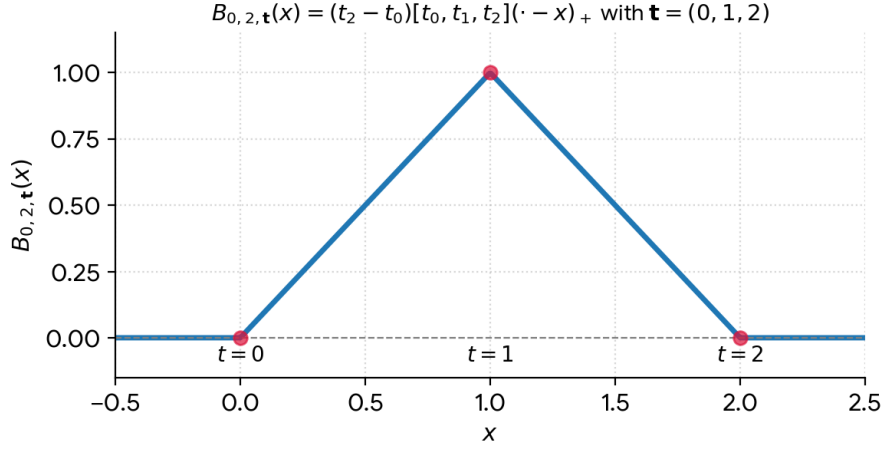


Figure 1: Example of a single B-spline.

1. B_i has small support, i.e. $B_i(x) = 0$ for all $x \notin [t_i, t_{i+k}]$.
2. $\sum_i B_i(x) = \sum_{i=r+1-k}^{s-1} B_i(x) = 1$ for all $t_r < x < t_s$.
3. B_i is strictly positive on its support, i.e. $B_i(x) > 0$ for $t_i < x < t_{i+k}$.

The above three properties show that the B-spline sequence $(B_i)_i$ consists of nonnegative functions which sum up to 1, meaning they provide a partition of unity.

As we see in the following definition, one can combine B-splines to construct a spline function.

Definition 1.3 (Spline function, and a collection of them). *A spline function of order k with knot sequence t is any linear combination of B-splines of order k for the knot sequence t . The collection of all such functions is denoted*

$$\mathcal{S}_{k,t} := \left\{ \sum_i \alpha_i B_{i,k,t} : \alpha_i \in \mathbb{R}, \text{ all } i \right\}.$$

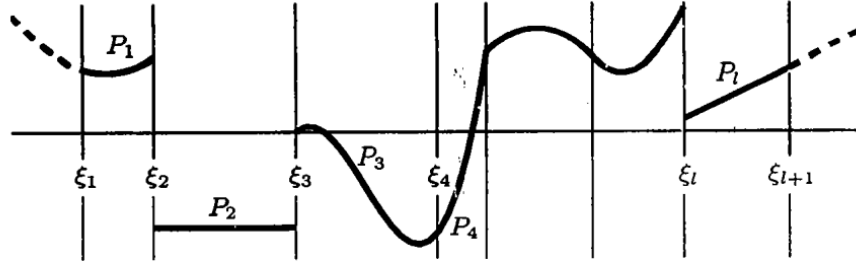
Note that for both exact representation of the piecewise polynomial functions and universal approximation of continuous functions up to error ϵ , we only need finite number of the splines, hence we never really sum to infinity here.

There is a theorem by Curry and Schoenberg (1966) which shows that a particular type of piecewise polynomial function space can be represented exactly by B-splines. First let us define this space.

Definition 1.4 (space of piecewise polynomials, $\mathbb{P}_{k,\xi}$). Let $\xi = (\xi_i)_{i=1}^{l+1}$ with $\xi_1 < \xi_2 < \dots < \xi_{l+1}$. $\mathbb{P}_{k,\xi}$ is the space of all functions f on $[\xi_1, \xi_{l+1}]$ that reduce to a polynomial of degree $< k$ on each subinterval (ξ_i, ξ_{i+1}) :

$$\mathbb{P}_{k,\xi} := \{f : [\xi_1, \xi_{l+1}] \rightarrow \mathbb{R} : f|_{(\xi_i, \xi_{i+1})} \in \mathbb{P}_k, i = 1, \dots, l\}.$$

(Here $\mathbb{P}_k$ denotes polynomials of order k , i.e., degree $< k$. Functions in this space can be completely discontinuous at the interior breakpoints $\xi_2, \dots, \xi_l$.)



(3) FIGURE. A pp function of some order, with $l = 7$.

Figure 2: Example of an element of $\mathbb{P}_{k,\xi}$

Definition 1.5 (space of piecewise polynomials with smoothness, $\mathbb{P}_{k,\xi,\nu}$). Let $\nu = (\nu_2, \dots, \nu_l)$ be an integer sequence with $0 \leq \nu_i \leq k$. $\mathbb{P}_{k,\xi,\nu}$ is the subspace of $\mathbb{P}_{k,\xi}$ satisfying $k - \nu_i$ continuity conditions (matched derivatives) across each interior breakpoint ξ_i :

$$\mathbb{P}_{k,\xi,\nu} := \{f \in \mathbb{P}_{k,\xi} \mid D^{j-1}f(\xi_i^+) = D^{j-1}f(\xi_i^-), \quad j = 1, \dots, k - \nu_i, \quad i = 2, \dots, l\}$$

Observe that each ν_i tells us the smoothness defect at each knot point ξ_i , and the smoothness there is precisely $C^{k-\nu_i-1}$.

Theorem 1 (Curry and Schoenberg). For a given strictly increasing sequence $\xi = (\xi_i)_{i=1}^{l+1}$, and a given nonnegative integer sequence $\nu = (\nu_i)_2^l$ with $\nu_i \leq k$, all i , set

$$n := k + \sum_{i=2}^l (k - \nu_i) = kl - \sum_{i=2}^l \nu_i = \dim \mathbb{P}_{k,\xi,\nu}$$

and let $\mathbf{t} := (t_i)_{i=1}^{n+k}$ be the nondecreasing knot sequence obtained from ξ by the following two requirements:

(i) for $i = 2, \dots, l$, the number ξ_i occurs exactly $k - \nu_i$ times in $\mathbf{t}$;

(ii) $t_1 \leq t_2 \leq \dots \leq t_k \leq \xi_1$ and $\xi_{l+1} \leq t_{n+1} \leq \dots \leq t_{n+k}$.

Then, the sequence $B_1, \dots, B_n$ of B-splines of order k for the knot sequence $\mathbf{t}$ is a basis for $\mathbb{P}_{k,\xi,\nu}$, considered as functions on $I_{k,\mathbf{t}} = [t_k \dots t_{n+1}]$. That is,

$$\mathcal{S}_{k,\mathbf{t}} = \mathbb{P}_{k,\xi,\nu} \quad \text{on } I_{k,\mathbf{t}}.$$

Proof. See de Boor (1978), Chapter IX, pp. 115. ■

Here, we can think of n as the total degrees of freedom, where kl is the total possible smoothness and $\sum_{i=2}^l \nu_i$ is the total smoothness constraints. This theorem provides a recipe for the construction of a B-spline basis for any particular piecewise polynomial space $\mathbb{P}_{k,\xi,\nu}$. Note that there must be k knots both before ξ_1 and after ξ_{l+1} . In practice, people usually set the k knots to be exactly at ξ_1 and ξ_{l+1} respectively. In addition, the standard construction would have each interior breakpoint ξ_i have $k - \nu_i$ knots.

Personally, I interpret each of the B-splines to be a sliding window of size n across the sequence of knots $[t_1, t_2, \dots, t_{n+k}]$.

1.1 Evaluation of B-splines

The following recurrence relation is useful and follows from an application of Leibniz formula to the original k^{th} divided difference definition of B-splines. Fixing a particular knot sequence $\mathbf{t}$,

$$B_{i,k}(x) = \frac{x - t_i}{t_{i+k-1} - t_i} B_{i,k-1}(x) + \frac{t_{i+k} - x}{t_{i+k} - t_{i+1}} B_{i+1,k-1}(x).$$

And so we can compute a particular B-spline $B_{i,k}(x)$ by repeatedly forming positive linear combinations of positive quantities, starting with the information from:

$$B_{j,1}(x) := \begin{cases} 1, & t_j \leq x < t_{j+1} \\ 0, & \text{otherwise} \end{cases}$$

Note that the above initialization and recurrence relation form an alternate definition of B-splines, though the properties of B-splines such as the smoothness properties

is not as clear is not as clear from this definition.

de Boor provides an algorithm to evaluate B-spline functions based on the above recurrence relation. It is a standard part of code packages and hence I do not describe it here.

1.2 Differentiation of B-splines

Since B-splines exactly represent piecewise polynomial (pp) functions, we can obtain their derivatives by accordingly differentiating pp functions. Doing so leads to the following expression,

$$D \left(\sum_{i=r}^s \alpha_i B_{i,k}(x) \right) = \sum_{i=r}^{s+1} (k-1) \frac{\alpha_i - \alpha_{i-1}}{t_{i+k-1} - t_i} B_{i,k-1}(x)$$

with $\alpha_{r-1} := \alpha_{s+1} := 0$.

So we see that the first derivative of a spline function can be found by simply differencing its B-spline coefficients.

Next, by repeated application of the above formula, the following is the j^{th} derivative of the spline function:

$$D^j \left(\sum_i \alpha_i B_{i,k}(x) \right) = \sum_i \alpha_i^{(j+1)} B_{i,k-j}(x) \quad \text{for } j < k$$

where the coefficients are computed recursively for $j > 0$ by:

$$\alpha_i^{(j+1)} := \begin{cases} \alpha_i, & j = 0 \\ (k-j) \frac{\alpha_i^{(j)} - \alpha_{i-1}^{(j)}}{t_{i+k-j} - t_i}, & j > 0 \end{cases}.$$

1.3 Integration of B-splines

Integration is given the following,

$$\int_{t_1}^x \sum_{i=1}^n \alpha_i B_{i,k}(y) dy = \sum_{i=1}^{s-1} \left(\sum_{j=1}^i \alpha_j \frac{t_{j+k} - t_j}{k} \right) B_{i,k+1}(x)$$

on $t_1 \leq x \leq t_s$.